Bézout and integer equations

Advanced high school · extension · Progressive exercises · 04

Exercise 1. Find $\gcd(84, 30)$ and a Bézout identity.

Exercise 2. Does $12x + 18y = 5$ have integer solutions?

Exercise 3. Find all integer solutions of $5x + 3y = 1$.

Exercise 4. Among solutions of $5x + 3y = 31$, find those with nonnegative x and y.